

Section B

Answer all the questions in the spaces provided.

2. A class of students were using dice to model radioactive decay. There were 10 groups of students in the class. Each group of students had 90 dice.



- (a) (i) Describe how they should carry out the experiment.

[3]

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- (ii) State the number of dice you would expect to land with a 6 facing up on the first throw in **each** group.

[1]

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- (iii) Each group's results were added together to give the class results. Give **one** advantage of using a bigger sample size.

[1]

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(b) The table shows the results for the **whole class**.

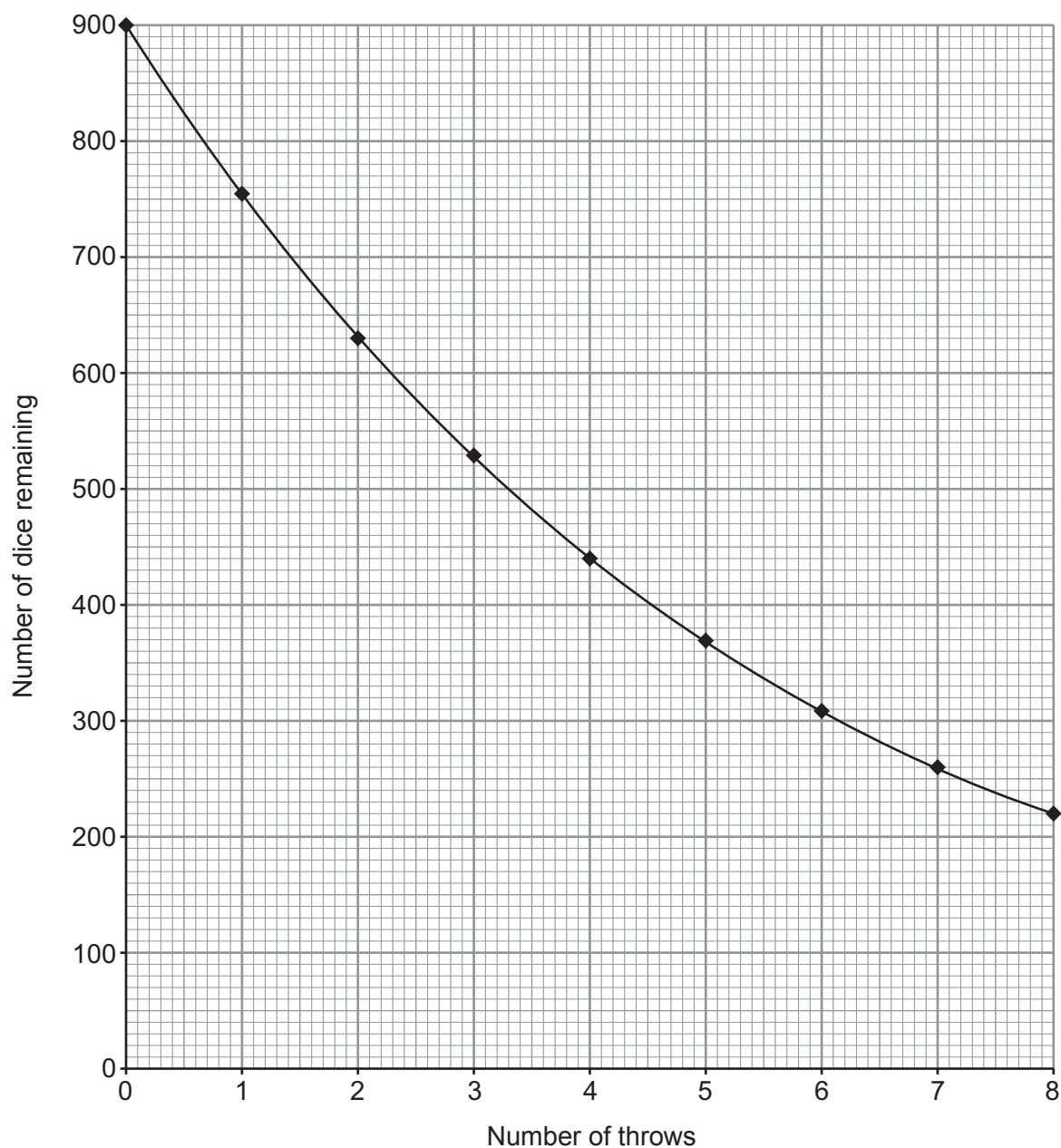
Throw	Number of sixes	Number of dice remaining
0	0	900
1	146	754
2	124	630
3	103	527
4		442
5	72	370
6	61	309
7	52	257
8	38	219
9	34	

(i) **Complete** the table. [2]

(ii) Use the data in the table to estimate the half-life of the 'dice'. [1]

half-life = throws

(c) Some of the data in the table is plotted on the grid below.



(i) **Draw lines** on your graph to find the half-life of the dice.

[2]

half-life = throws

(ii) State **one** reason why finding the half-life using the graph is a better method than the one used in part (b)(ii).

[1]

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- (d) Radioisotopes have many medical uses. One use is to sterilise blood products for transfusion.

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only



- (i) Select from the table below the most suitable radioisotope for this purpose. [1]

Radioisotope	Half-life	Type of emissions
bismuth-213	46 minutes	alpha
chromium-51	28 days	gamma
caesium-137	30 years	beta and gamma
bismuth-209	2×10^{19} years	alpha

- (ii) Give **two** reasons for your choice. [2]

1.
2.

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